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Mapping entrepreneurial inclusion across US neighborhoods: The case of low-code e-commerce entrepreneurship

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Abstract

Entrepreneurship is presented as a path to prosper through commerce, yet there is evidence that certain communities are underrepresented in entrepreneurship. Technological developments such as low-code e-commerce tools have altered the barriers to launch commercial ventures. Do these tools alter entrepreneurial inclusion? We study an omnipresent low-code tool, Shopify, which has dramatically reduced the financial and technical barriers to e-commerce. We undertake an abductive analysis using unique data on the spatial distribution of Shopify-based activity for the entire United States. We find evidence of entrepreneurial inclusion; neighborhoods with more Black residents, a group historically underrepresented in entrepreneurship, also have more low-code ventures. We interpret these results against traditional forms of startups: newly registered businesses and VC-backed e-commerce ventures. These benchmark analyses further support our findings.

Managerial Summary: We consider the impact of recent technological changes that make it easier to start entrepreneurial ventures. We study the spatial uptake of a popular low-code e-commerce tool, Shopify, that makes launching and running an online business

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markedly cheaper and easier. Using data on nearly 160,000 Shopify-based ventures across the entire United States, we find that usage of Shopify is more prevalent in neighborhoods with more Black residents. The pattern is distinct from that observed for traditional ventures and VC-backed start-ups. The evidence suggests low-code tools represent a novel path to entrepreneurship, enhancing entrepreneurial access and potentially opening previously under-explored markets. The finding further highlights the role of new tools (e.g., low-code or AI-driven) as a possible addition to policies aimed at enhancing entrepreneurial inclusion.

KEYWORDS

entrepreneurship, geography, inclusion, low-code tools, neighborhoods

1 | INTRODUCTION

Entrepreneurial ventures are espoused as an important source of economic growth, both for the individuals who pursue them and for society (Schumpeter, 1942). Scholars have studied who entrepreneurs are (e.g., background); what they do (e.g., industry); how they do it (e.g., the underlying technology); and where they do it (e.g., geography). And while extant work offers keen insights into these questions, it is also the case that entrepreneurial activity has dramatically changed since it first became a topic of academic scrutiny. The advent of the Internet, for example, changed the face of commerce as technology-savvy individuals—based disproportionately in places like Silicon Valley—built and scaled entrepreneurial ventures that cater to large markets in the United States and beyond.

Recent technological developments have shifted the barriers to launch commercial ventures. In the past, the development of new ventures necessitated substantial resources. E-commerce ventures, for example, proliferated in communities where tech-savvy individuals could raise large sums of money to hire talented engineers, buy the necessary computing equipment, and pay for offices to accommodate them (Feldman, 2001; Florida & Kenney, 1988; Saxenian, 1996; Sorenson & Audia, 2000; Vedula & Fitza, 2019). We call attention to and contrast this with the advent of low-code e-commerce tools, such as Shopify, that afford ready-made technical tools to support robust e-commerce operations across a wide number of countries and currencies (Cutolo & Kenney, 2021). Utilizing such tools does not require technical know-how. There is also no need for a substantial upfront investment, allowing one to launch an e-commerce business with as little as a few dozen dollars per month and reach national or even international customers (Dushnitsky & Stroube, 2021). This shift in the techno-economic profile of new ventures may have consequences for *where* such ventures can conceivably be started and operated.

The purpose of this study is to explore how this changing techno-economic profile shapes broader patterns of entrepreneurial access and the inclusion of historically underrepresented groups. To understand the impact of low-code e-commerce tools, we undertake a large-scale

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investigation of where Shopify-based ventures are geographically located (Figure 1). Notably, our investigation opens avenues for future work more broadly by highlighting a little-studied yet common phenomenon that poses opportunities to advance our understanding of entrepreneurial activity (Benner & Waldfogel, 2023; Clough et al., 2019; Hallen et al., 2020).

Our study is guided by an abductive approach; that is, a generative process that starts with an anomaly and proceeds to create and evaluate traditional and novel explanations that render it understandable (Behfar & Okhuysen, 2018; Gelman & Imbens, 2013; Sætre & Van de Ven, 2021). Specifically, we are interested in the use of low-code e-commerce tools as reflecting a "novel" path to entrepreneurship; that is, entrepreneurship that need not be heavily dependent on the local endowments of capital (be that the level of human, financial, or social capital in the focal community). Our investigation centers on the relationship between the spatial distribution of entrepreneurship and the demographic composition of neighborhoods; in particular, the prevalence of the Black population, a group historically underrepresented in entrepreneurship.

The investigation is facilitated through a major data effort. We construct a detailed dataset of the demographics of the 32,000 neighborhood ZIP Code Tabulation Areas (ZCTAs) in the United States and integrate it with information on low-code e-commerce ventures in those neighborhoods. We utilize information about technologies underlying e-commerce websites to systematically identify nearly 160,000 e-commerce ventures operating all over the United States that use a popular low-code tool, Shopify. A neighborhood-level analysis is particularly useful to understand the techno-economic changes to entrepreneurial access and its spatial distribution. First, the analysis captures rich cross-neighborhood variation and thus goes beyond aggregate state-level variation, which has been used to study institutional-economic drivers of entrepreneurial access (i.e., the study of cross-state top-down policy and regulatory differences).

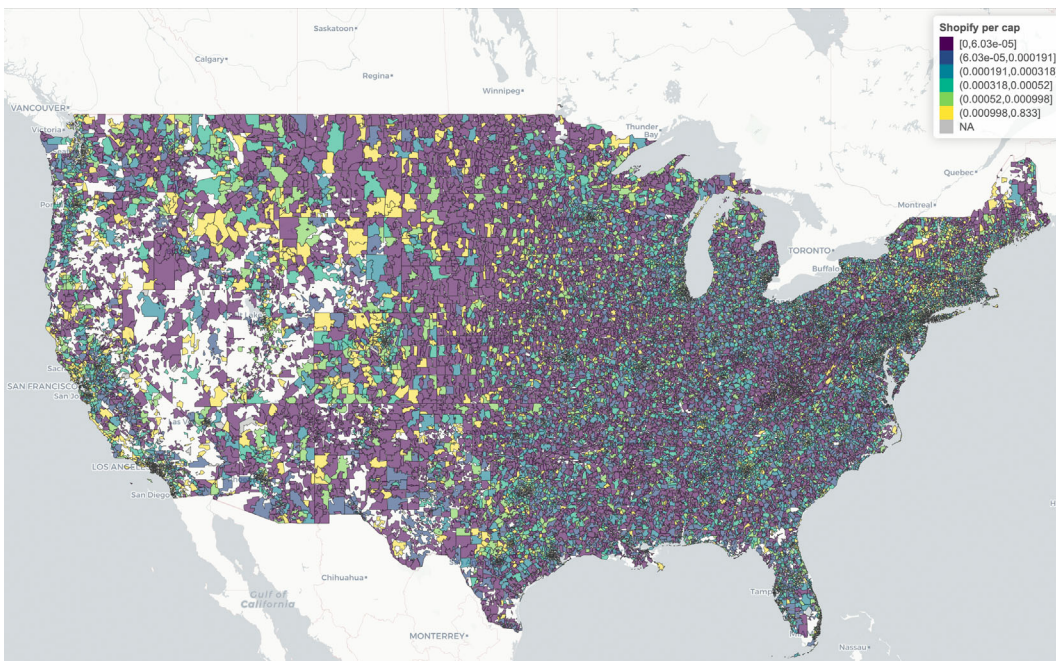

FIGURE 1 Density of Shopify ventures at the ZCTA level.

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Second, we study the attributes of the specific communities in which the ventures are embedded, which sheds light on the issue of entrepreneurial access (Gagliardi & Sorenson, 2023). To see that, simply note that the greater Boston metropolitan area is lauded as a startup hub, yet there is significant startup variance within this area: for example, the differences in startup activity between Cambridge and South Boston (Sorenson et al., 2016).

The neighborhood-level abductive analysis yields several insights. First, we shed light on the factors associated with the distribution of Shopify businesses across the United States. Our main finding is that neighborhoods with more Black residents are home to more Shopify-based ventures. These patterns persist in the face of traditional explanations of the spatial distribution of entrepreneurial activity: that is, the endowment of local capital.

Second, the patterns are unique to Shopify-based ventures and stand out in comparison to traditionally studied forms of entrepreneurial activity. Specifically, we examine the same patterns for two common measures of entrepreneurship: a broader measure of new business creation (regardless of sector) that encompasses millions of ventures, and a narrower measure of VC-backed e-commerce ventures that encompasses a few hundred ventures. These comparisons provide additional strength to the interpretation that Shopify enables a unique type of entrepreneurship that appears to be more inclusive of otherwise underrepresented populations.

The following section outlines the context for our study: low-code e-commerce tools (Section 2). We then review the disparity in, and extant explanations of, the distribution of entrepreneurial activity (Section 3). Section 4 details how we created the public integrated dataset of all the neighborhoods in the United States and Shopify-based ventures therein. Section 5 presents our analysis of the spatial distribution of those ventures using maps and regression analyses. Section 6 reports further empirical tests that help to theoretically situate our main result by using benchmark comparisons with other types of entrepreneurship and subsample analyses. Finally, we expand on potential theoretical mechanisms and discuss the implications of these findings for our understanding of entrepreneurial access (Section 7).

2 | AN ABDUCTIVE STUDY OF LOW-CODE E-COMMERCE TOOLS AND ENTREPRENEURIAL ACCESS

Entrepreneurship is celebrated as a path to prosper through commerce (Rindova et al., 2009). Yet, access to entrepreneurship is not equally distributed across the population. Entrepreneurs from some demographic groups, such as women or Blacks or African Americans, are underrepresented (Fairlie & Robb, 2008; Gompers & Wang, 2017). However, technological developments have led to new tools that have altered the relative levels of entrepreneurial representation within a population. For example, in the late 20th century, personal computers (PCs) became an omnipresent digital tool, and access to PCs was associated with a greater likelihood of women launching a business venture (Fairlie, 2006).

This article seeks to inform the relationship between low-code tools and entrepreneurial access. To that end, we undertake an abductive approach (Behfar & Okhuysen, 2018; Gelman & Imbens, 2013; Mantere & Ketokivi, 2013; Sætre & Van de Ven, 2021). Abductive reasoning does not deductively validate a preexisting general theory. Nor does it inductively build a new general theory. Rather, it is a process by which a scholar evaluates a set of plausible explanations (Gelman & Imbens, 2013; Sætre & Van de Ven, 2021). This study follows the abductive

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approach. Our focus is on a "novel anomaly" that may reshape entrepreneurial inclusion; namely, the advent of low-code e-commerce tools (e.g., those available through Shopify) as a viable path to entrepreneurship. Shopify also meets the definition of a "novel phenomenon" given that the academic literature does not yet have accepted definitions or shared language for talking about the phenomenon, and it is not clear which research communities might most benefit from its study (O'Mahony & Cohen, 2022).

Therefore, in this section, we introduce our context across two dimensions. We first discuss the features of the novel tool in light of factors that are theoretically known to affect launching new ventures (Section 2.1). We further explain how the entrepreneurship facilitated by these tools fundamentally differs from that facilitated by another novel phenomenon, namely marketplaces such as eBay or Amazon (Section 2.2).

2.1 | The features of low-code e-commerce tools

We study the e-commerce vertical, a context where goods and services are sold via the Internet. E-commerce activity has grown dramatically since the advent of the Internet in the 1990s (Anderson, 2006; Brynjolfsson & Smith, 2000; Oestreicher-Singer & Sundararajan, 2012). The vertical has been a hotbed of entrepreneurial activity, with Diappers.com founded by Marc Lore in the 1990s, Bonobos launched by Andy Dunn and Brian Spaly in the 2000s, and Direct To Consumer ventures such as DollarShaveClub (by Michael Dubin and Mark Levine) and Warby Parker (by David Gilboa, Jeffrey Raider, and co-founders) in the 2010s. By 2019, about 12%–15% of all retail activity in the United States was conducted online.[1]

An important feature of all e-commerce businesses is that they are not limited to geographically proximate consumers and often cater to customers located across the nation or even internationally. This attribute marks a departure from an assumption in traditional studies of entrepreneurial activity; that is, that businesses set up by self-employed or budding entrepreneurs predominantly serve their local community (Robb & Farhat, 2013). We iterate this observation to highlight that Shopify-based businesses are not limited to geographically local demand and can cater to like-minded customers in their immediate neighborhoods as well as those further away.[2]

Given these general features of e-commerce activity, the focus of this article is a new type of technological tool that constitutes a "novel anomaly" in e-commerce entrepreneurship. Our observation is informed by recent work on the effects of digitization on tools and activity in this vertical (Cutolo & Kenney, 2021; Dushnitsky & Stroube, 2021). Some tools cater to a specific function of an e-commerce business (e.g., Gusto, Klaviyo, Notion, Twilio) and others support a broader set of functions (e.g., Magento, Shopify, Wix). As shown below, the advent of these low-code tools raises questions regarding common assumptions about the technical and financial backgrounds that have, to date, characterized founders of successful e-commerce businesses.

[1]Federal Reserve Economic Data: E-Commerce Retail Sales as a Percent of Total Sales (ECOMPCTSA), reported at https://fred.stlouisfed.org/series/ECOMPCTSA/.

[2]Recent work offers excellent analysis of a mirror question, focusing on customers' rather than ventures' locations. For example, Pongeluppe (2022) explores how spatial inequality perpetuates socioeconomic disparity in consumer access to goods or services. Our focus is the entrepreneurs who launch new goods or services. Specifically, we study spatial distribution in entrepreneurial activity as indicator of entrepreneurs' socioeconomic background.

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To facilitate the discussion, we will focus on Shopify, a popular e-commerce low-code tool. Shopify tools are used by the largest number of e-commerce web domains in the United States. Data on e-commerce tools find that Shopify's fraction is bigger than the combined fraction of its major competitors (i.e., Magento, Wix, and WordPress's WooCommerce).[3] Moreover, industry reports suggest Shopify is particularly strong among SMEs, which is the focus of our study.[4]

We proceed to highlight the features of low-code tools that may affect entrepreneurial access. These features relax conventional assumptions about (i) the necessity of technical know-how, (ii) the required initial financial capital, and (iii) the ability to scale and engage in meaningful growth. First, the use of a low-code tool eliminates the need for significant technical know-how. Shopify and similar tools cater to the layperson—those with human capital but not necessarily technical backgrounds. They enable the construction of a functioning e-commerce business through a friendly drag-and-drop interface that is accessible to those with business acumen yet little background in coding or technology development. Anecdotal evidence to that effect is illustrated by the quote below;

> Taiwo Adegbulu, who co-founded Omolola Jewellery with her sister in 2018, says they chose Shopify to launch their business because its ease of use allowed them to build a website without any prior experience.[5]

Second, there is no need for substantial upfront investment to launch a business. Shopify charges a relatively low monthly fee (under $50 per month as of 2024) to access tools that are needed to launch an e-commerce business. It therefore changes the financial needs of a budding e-commerce entrepreneur, transforming the need for a large upfront capital expenditure to build e-commerce operations (e.g., hiring engineers, buying servers, renting offices) into a small variable cost that is paid as part of the ongoing business operational expenses. As a result of this fundamental change, ventures that previously might not have been launched, or would require substantial VC funding, are now possible. Adegbulu noted of her Shopify venture, "*We started with £200*."[6] The insight is echoed by Aaron Levie, the CEO and co-founder of Box, who is knowledgeable about launching and scaling highly successful ventures using external funding. Levie's comment reflects the fundamental change associated with low code tools:

> I have a friend who...sells balloons online and, and I don't think he would have started the business if Shopify didn't exist. Like, I think, like, the existence of Shopify made it so he could, like, be like, oh, well, I had this, like, random idea, let's just see if it works. It lowered the barrier to then going out and basically

[3]The information is based on BuiltWith data on the adoption of low-code tools across the all US-based e-commerce web domains in the year 2019. Section 4.1 reviews the BuiltWith database and the decision to focus on 2019.

[4]This is corroborated through other sources, such as an independent ecommerce advisory (https://www.junglescout.com/blog/best-ecommerce-software/), as well as one of its main competitors which acknowledges and ranks Shopify as the second "best e-commerce platform" (https://www.bigcommerce.co.uk/articles/ecommerce/ecommerce-platforms/).

[5]Taiwo Adegbulu shared her story at the SMS London 2022 conference. The specific quotes appear in a Performance Marketing article, accessible at https://www.performancemarketingworld.com/article/1739903/small-businesses-used-shopify-rapidly-scale.

[6]Daily Express article, accessible at https://www.express.co.uk/finance/personalfinance/1468204/business-small-set-up-side-hustle-entrepreneur-tips-exclusive.

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starting a business. (Aaron Levie on the May 22, 2024 episode of The Twenty Minute VC; 42:51 mark)[7]

Third, and relatedly, one can use low-code tools to seamlessly scale up operations without dramatically changing operations. For example, Shopify offers access to additional low-code tools that are necessary as e-commerce activity expands in volume, across different countries, or into multiple currencies. It follows that entrepreneurs using Shopify tools are associated with a lower need for initial financial capital or subsequent human capital compared to peer e-commerce ventures (Dushnitsky & Stroube, 2021). In sum, low-code tools mark a different path into launching and growing e-commerce ventures.

2.2 | E-commerce businesses versus businesses on e-commerce marketplaces

We now turn to the important distinction between (a) the focus of this article: e-commerce businesses built using low-code tools, and (b) businesses that sell on e-commerce marketplaces such as Amazon Marketplace, eBay, or Etsy.

Shopify is a technology for building and operating an e-commerce business, similar to other technologies such as WordPress, Wix, and BigCommerce. Conceptually, Shopify-based ventures represent standalone entrepreneurial businesses. Creating a Shopify-based venture entails (i) market creation and management of all facets of the business with (ii) a substantial growth potential. On the most practical level, the use of Shopify is invisible to the customers of such businesses; that is, a customer purchases a product or service on a dedicated business website. Put simply, when visiting the website of a Shopify venture, the customer is offered only the products and services of that venture.

In contrast, multi-sided marketplaces that connect merchants to customers (e.g., Amazon Marketplace, eBay, Etsy) are effectively responsible for sales and distribution. Rather than purchasing on a business' dedicated website, customers visit the marketplace website. Once on that marketplace website, the customer can search for products, and the search yields a webpage listing similar products across different businesses. This relationship with the customer is also reflected in taxation. Because e-commerce marketplaces manage the relationship with the customer, they are responsible for collecting sales tax. In contrast, a Shopify-based venture manages its own relationship with its customers and is therefore responsible for collecting and reporting sales tax.[8]

From the entrepreneurs' perspective, the advantage of reaching a broad mainstream customer audience on a marketplace comes with a notable disadvantage; they become subject to a significant power imbalance. This is because entrepreneurs must conform to the algorithms, processes, and policies of a marketplace in order to sell on it (Cutolo & Kenney, 2021). Moreover, the risks of this dependence increase as those selling through a marketplace attempt to scale up. Marketplaces may implicitly favor some entrepreneurs, engendering rich-get-richer effects that disproportionately benefit a few businesses with superstar offerings while effectively shifting sales away from less mainstream businesses (Brynjolfsson et al., 2010; Tan et al., 2017).

[7]https://www.thetwentyminutevc.com/aaron-levie-2/.
[8]https://www.a2xaccounting.com/ecommerce-accounting-hub/us-sales-tax-shopify.

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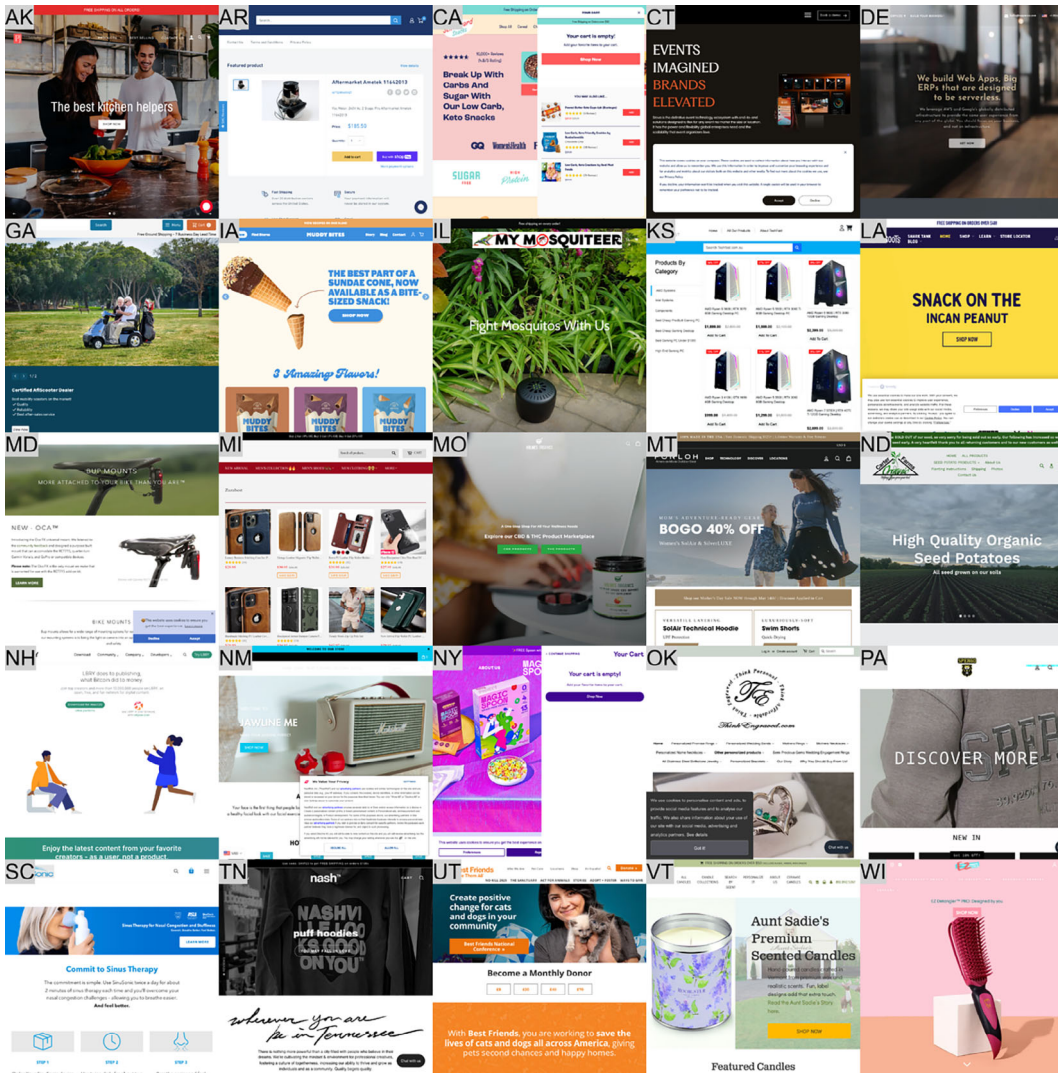

FIGURE 2 Illustration of popular Shopify ventures from the sample. One venture for 25 different states, ordered from AK (Alaska) to WI (Wisconsin); the top left corner of each image denotes the state. Popularity based on web traffic or page rank. Includes those launched in 2018 or later.

Businesses that sell on marketplaces may even face competition from the marketplace itself (Toh & Agarwal, 2023; Zhu & Liu, 2018).

Note that Shopify provides access to low code tools but does not serve as a marketplace. Hence, launching a venture using Shopify does not result in the same dependence dynamics because the use of Shopify is largely invisible to customers, and a business can retain its relationship with customers while migrating between various e-commerce backends if desired.[9] As

[9]For example, Shopify competitor Wix has an article in its help center titled "Migrating from Shopify to Wix Stores" (https://support.wix.com/en/article/wix-stores-migrating-from-shopify-to-wix-stores). And Shopify has an article in its help center instructing on how to move in the opposite direction: "Migrate from Wix" (https://help.shopify.com/en/manual/migrating-to-shopify/migrating-from-wix).

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with traditional entrepreneurial ventures, the ultimate growth of a Shopify venture is a function of its ability to engage in entrepreneurial actions such as market creation and scaling. Oftentimes, these ventures target untapped markets rather than the mainstream customers that search on popular marketplaces.

To avoid confusion, and given these differences, we explicitly note that our focus is on e-commerce businesses rather than businesses on e-commerce marketplaces.[10] That is, we study Shopify-based ventures which are e-commerce ventures built using low-code tools. A visual corollary of the difference of these two types of e-commerce entrepreneurship is illustrated in Figure 2. Customers who buy from Shopify ventures search for and engage directly with those businesses, whereas customers who buy on a marketplace (e.g., Amazon or Etsy) are limited to search within the marketplace and usually engage with the marketplace rather than the underlying businesses. Figure 2 highlights the unique visual diversity of the Shopify businesses in our sample. In contrast, customers of businesses selling on Amazon Marketplace engage with such ventures through Amazon's standardized templates.

In sum, we underscore features of low-code tools that cast a new path to entrepreneurship, a path that mitigates the dependence on human, financial, and social capital, and that is different from both traditional e-commerce entrepreneurship (per Section 2.1) and marketplaces for e-commerce activity (per Section 2.2). To the extent that is the case, entrepreneurship may be common in areas and communities that have been traditionally underrepresented. To facilitate the investigation, we proceed to review the extent of disparity in entrepreneurial activity as documented by past studies.

3 | REVIEW OF RACE, GEOGRAPHY, AND ENTREPRENEURSHIP

A large body of research explores patterns of entrepreneurial activity by geography and race. The work documents unequal access by different racial groups (Bruton et al., 2023). We review recent studies that address the question of representation in entrepreneurship, motivated by the well-documented Black/White racial divide across various aspects of life in the United States (Ruef, 2022; Ruef & Grigoryeva, 2018). One of the most salient features of the divide is evident in the distinct racial profiles of neighborhoods across the majority of US cities (Charles, 2003). To inform our investigation, we briefly review research on disparities from the perspectives of (1) its magnitude on entrepreneurship and (2) its relationship to geography.

First, there is substantial evidence that the Black population in the United States is underrepresented in entrepreneurship. Disparities are observed for self-employment, business ownership, and VC-backed ventures. Throughout the twentieth century, the rates of self-employment for Black men hovered at around 30%–40% the rates of self-employment for White men (Fairlie & Robb, 2008, Figure 2.2). This pattern of disparity is more evident when it comes to business ownership in general and ownership of innovation-intensive businesses in particular. Blacks have an average business ownership rate of 4.4%, or about half of the 10.1% for Whites (Fairlie & Robinson, 2022). The disparity is exacerbated among innovation-intensive firms, where the rate for Blacks is about a third that of Whites: 1.1% versus 3%, respectively (Fairlie &

[10]Both Shopify and companies like eBay are sometimes referred to as a "platforms." We hope this section highlights that they are fundamentally different types of platforms: Shopify is more usefully characterized as a technological platform, while eBay a marketplace platform (Gawer, 2014).

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Robinson, 2022). Similarly, Gompers and Wang (2017) note that less than 1% of all founders of venture-capital-backed startups are Black.

Second, extant work suggests the racial disparities in entrepreneurship are related to geographic location. Communities with a large fraction of Black residents exhibit lower rates of entrepreneurship (Fairlie & Robb, 2008). Moreover, racial segregation itself has been found to suppress entrepreneurship (Fairchild, 2008; Fischer & Massey, 2000). For example, Samila and Sorenson (2017) find that racially segregated cities receive fewer benefits from venture capital investment than more integrated cities. Prior studies point to (the lack of) local capital endowments as an explanation of the spatial disparity in entrepreneurial access (Vedula & Fitza, 2019). Faber (2019) argues that access to financial services—which is necessary for entrepreneurial activity—is not randomly distributed with respect to the racial composition of neighborhoods. Kwon et al. (2013) focus on another capital endowment: local social capital. They observe that variation in social capital across communities is one such geographic mechanism that can affect racial disparities in entrepreneurship. Taken together, the evidence suggests that entrepreneurial access shares the general association of entrepreneurial activity with the local endowment of human, financial, and social capital (Feldman, 2001; Florida & Kenney, 1988; Klepper, 2010; Saxenian, 1996; Sorenson & Audia, 2000).

4 | DATA AND SAMPLE

We construct a unique database of the geographic distribution of Shopify ventures at a neighborhood level. The first step in this process was identifying where Shopify-based ventures were located. We study Shopify's largest geographic market, the United States. The advantage of focusing on neighborhoods rather than more aggregate geographies is threefold. First, it allows us to control for regulatory effects. The cross-neighborhood analysis captures fine-grained variation that goes beyond regulatory or policy differences usually dictated at the state or municipal level. Second, and related, our approach avoids the problem associated with analysis of aggregate territories such as MSAs that can span the policy regimes of multiple states (e.g., the MSA covering New York City also spans parts of New Jersey and Pennsylvania; the MSA covering Washington, DC also spans pars of Maryland, Virginia, and West Virginia). Third, the neighborhood-level analysis is fully aligned with the theoretical community-level mechanisms explored in the literature. These data offer insights into entrepreneurial access by studying the attributes of the specific community in which entrepreneurs are embedded. To see that, note that greater Boston metropolitan is lauded as a startup hub, yet there are fewer startups that are based in South Boston (Sorenson et al., 2016). Analysis at the State or MSA level can miss out on features of entrepreneurs' immediate community (e.g., a study of the Boston metropolitan area blends features of Cambridge and South Boston despite meaningful differences between these areas). Taken together, a fine-grained neighborhood-level analysis is particularly appropriate for empirically capturing the theoretical impact of low-code tools on entrepreneurial access and underrepresented communities.

4.1 | Measuring the geographic distribution of Shopify-based entrepreneurial activity

The data on US ventures using Shopify was downloaded from the web-technology search engine BuiltWith. BuiltWith has been increasingly employed in academic work that measures

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technology usage (Stroube & Dushnitsky, 2023). BuiltWith indexes the technology-usage profile of public web domains much in the same way Google indexes the readable text of public websites: it allows one to search the internet for web domains that use a specific technology, as well as to view what technologies a specific web domain uses (for recent academic use, see Burford et al., 2022; Hallen et al., 2023). Shopify-based ventures, by their very nature, are identifiable based on their underlying web technologies. As a result, we can construct a comprehensive dataset of all relevant ventures.

An advantage of our approach is that it identifies e-commerce ventures without the need to resort to administrative data or regulatory records. This alleviates several common concerns in the entrepreneurship literature regarding sample selection based on venture attributes or scale (e.g., over-sampling of public companies versus under-sampling of private ventures, or missing ventures below a minimum number of employees). The result is that we systematically observe entrepreneurial activity by even the most nascent of ventures. Our approach further mitigates concerns of self-report in surveys of entrepreneurs or administrative data from the providers of the tools that they utilize. By capturing the web technologies that underlie e-commerce ventures, we gain an objective account of the number of Shopify-based ventures.

In addition to the technology-use data, BuiltWith also provides metadata about the web domains, including geographic location data. This allowed us to target our BuiltWith search to domains that met the following criteria: (1) domains that were documented as using Shopify, and (2) domains that BuiltWith flagged as being in the United States. We study any domain that had used Shopify at some point during 2019—the most recent year prior to the pandemic. We filtered out any ventures with potentially geographically misleading ZIP Codes, such as ZIP Codes associated with PO Boxes. This generated a sample of 159,663 domains that comprise the total count of ventures using Shopify in our study.

To illustrate the nature and scope of Shopify ventures in our sample, Figure 2 presents a collage of 25 screenshots. Each screenshot is of a web domain in our sample that was first detected in 2018 or 2019 and each rank as one of the most popular (based on web traffic or page rank data) in its respective state. We note two observations about this figure. The first is its visual diversity. Each store has its own distinct brand and identity. In fact, most visitors to these websites are unlikely to even know they are Shopify-based ventures. The second observation is the diversity of products and services, which range from fashion and beauty to food and technology. This mirrors the broader e-commerce industry where different e-commerce businesses sell a diverse set of products and services.

4.2 | Location data: Validation and measure construction

BuiltWith systematically indexes the technology underlying web domains. The BuiltWith geographic location data used for this study is novel, and therefore we engaged in additional scrutiny and triangulation. This is because the technology data are directly detected from the underlying standardized web code, whereas location data are scraped by BuiltWith from the human-readable text on each web domain. We wanted to confirm that the geographic data provided by BuiltWith matched the actual venture locations. To validate the address data, a team of research assistants manually visited a subset of the domains. Research assistants independently searched for ZIP Codes at 5850 different domains by manually visiting each domain, searching for venture location data (i.e., addresses), and saving a screenshot of where they found the address information. In 75% of cases, the ZIP Codes provided by BuiltWith matched

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those found by at least one research assistant.[11] Thus, the BuiltWith data appear to accurately capture the location of the ventures.

Because we are interested in the geographic distribution of these ventures, we next aggregated counts of Shopify ventures to the unit of the ZCTA. ZCTAs are geographically defined areas that roughly correspond with postal ZIP Codes and are used by the Census Bureau as an official unit of geographic analysis. There are roughly 32,000 ZCTAs spread across the United States, each with an average population of approximately 10,000 people. Following prior research (Fazio et al., 2021), we used the UDS Mapper crosswalk table to map a venture's postal ZIP Code to a specific ZCTA.[12] The dependent variable is a count of Shopify ventures in each ZCTA.

4.3 | Neighborhood demographic variables

In line with past entrepreneurship research on the predictors of local entrepreneurial activity (e.g., Armington & Acs, 2002; Feldman, 2001; Florida & Kenney, 1988; Klepper, 2010; Saxenian, 1996; Sorenson & Audia, 2000; Vedula & Fitza, 2019), we assembled data on local endowments of human, financial, and social capital at the ZCTA level. We draw on US Census Bureau data in line with prior work (Fairlie & Robb, 2008; Fairlie & Robinson, 2022; Gompers & Wang, 2017). Specifically, we collected neighborhood-level demographic characteristics at the ZCTA level from the Census American Community Survey 2019 5-year estimates. These variables include counts of the (1) total population, (2) Black or African American population, (3) population with bachelor's degrees, and (4) total poverty: people aged 20–64 that were below poverty in the last 12 months.[13] Finally, in line with the social capital arguments of Putnam (2000), we also employed a measure of neighborhood social capital—count of religious, civic, and social organizations—which was from the National Neighborhood Data Archive. Tables 1 and 2 report summary statistics and correlations for these variables.

TABLE 1 Summary statistics.

	N	Mean	Std. dev.	Min	Max
Shopify count	32,647	4.89	16.88	0	719
Black/AA pop.	32,647	1397.08	4097.38	0	87,613
Total pop.	32,647	9945.24	14707.75	1	128,294
Bachelor degrees pop.	32,647	2172.28	3971.95	0	47,022
Below poverty pop.	32,647	709.97	1331.40	0	18,922
Social orgs count	32,647	17.96	26.36	0	427
All entrepreneurial businesses count	32,647	297.01	697.68	0	20,103
VC e-commerce businesses	32,647	0.01	0.12	0	7

[11]The research assistants retrieved location information for over half of the web domains they searched for. Inter-coder reliability stands at a high 95%.

[12]We dropped the roughly 350 ZCTA that the Census reported as having zero population.

[13]Given the digital nature of the ventures we are studying, we also considered using a count of population with internet subscriptions. However, we did not use it because of the high correlation with population (97%).

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TABLE 2 Correlation table.

	Shopify count	Black/ AA pop.	Total pop.	Bachelor degrees pop.	Below poverty pop.	Social orgs count	All entrepreneurial businesses count	VC e-commerce businesses
Shopify count	1.00							
Black/AA pop.	0.19	1.00						
Total pop.	0.43	0.59	1.00					
Bachelor degrees pop.	0.53	0.39	0.84	1.00				
Below poverty pop.	0.33	0.64	0.81	0.53	1.00			
Social orgs count	0.45	0.66	0.85	0.73	0.76	1.00		
All entrepreneurial businesses count	0.60	0.35	0.65	0.72	0.46	0.66	1.00	
VC e-commerce businesses	0.46	0.02	0.08	0.17	0.06	0.12	0.19	1.00

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4.4 | Validation of neighborhood demographics as proxy for venture demographics

Our interest is the geographic distribution of entrepreneurship, so it is helpful to clarify the relationship between neighborhood demographics and the potential founder demographics. We engaged in additional effort to validate that the demographic profiles of a neighborhood are broadly correlated with the demographic profiles of the founders of Shopify ventures located in those neighborhoods. To that end, a group of research assistants collected information on the identity of founders directly from a subset of domains in our sample. The subset of domains was created using a two-step approach. First, we identified the ZCTAs with the highest level of Shopify activity (i.e., 3017 ZCTAs each with >13 domains). Second, we filtered these ZCTAs by the percentage of the total population that was Black. We focused on ZCTAs in the top decile (Blacks account for 32% of the population) and bottom decile (Blacks account for 1.6%). There are 19,086 Shopify ventures in these deciles, from which we randomly selected 1450 ventures from the ZCTAs in the top decile and 1450 domains from those in the bottom decile.

The research assistants then combed through these 2900 Shopify ventures. Information about founders was available on "About Us," "Company History," "Our Story," or similar pages of a domain.[14] For the founders of domains in the top decile, approximately 32% were Black or African American; for the founders in the bottom decile, less than 1% were Black or African American. These numbers closely mirror the Census data on the proportion of Blacks in those ZCTAs (as aforementioned; 32% and 1.6%, respectively), thus suggesting neighborhood demographics proxy those of Shopify founders.

5 | EMPIRICAL DESIGN AND MAIN RESULTS

We now turn to the empirical patterns that can inform our investigation. Figure 1 presents a map of the contiguous United States where each ZCTA is color-coded based on the prevalence of Shopify ventures. Colloquially, the figure answers the question: "Where is Shopify Valley?" Three observations are noteworthy. First, the magnitude of low-code based entrepreneurship is substantial. One out of every two neighborhoods is home to at least one Shopify venture. Second, there is significant intra-state variation in the number of Shopify ventures. Methodologically, this observation illustrates the need for a fine-grained level of analysis that explores entrepreneurial activity at a local rather than state or metropolitan level. Conceptually, these patterns suggest that Shopify-based entrepreneurial activity mirrors community characteristics. Which leads to the third observation, namely, that the pattern of Shopify ventures does not seem to mimic any single neighborhood characteristic.

To elucidate this point, in the Online Appendix we present the spatial distribution of the abovementioned factors. Figure A.1 maps the fraction of the local population that is Black. Figures A.2–A.4 map the human, financial, and social characteristics of ZCTA areas, respectively. Glancing at these figures, there is no evidence of a clear correlation with the prevalence of Shopify ventures. A few words of caution are needed here. The ZCTA areas are not of similar size, and in fact, many ZCTA may be too small to clearly appear in a map of the contiguous United States. As a result, a visual inspection of the maps may offer only limited insight into the drivers of low-code based entrepreneurship.

[14]LinkedIn search yielded limited matches. This is likely because of the nature of many of the ventures under study.

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TABLE 3 Main results predicting the likelihood of Shopify ventures within ZCTAs.

Dependent variable	(1) Log (Shopify count +1)	(2) Binary outcome: Shopify >0	(3) Log (Shopify count +1)
Black/AA pop.	0.0307	0.0066	0.0141
	(0.0000)	(0.0030)	(0.0177)
Total pop.	−0.0905	0.0257	−0.3247
	(0.0087)	(0.0000)	(0.0000)
Bachelor degrees pop.	0.1431	0.0509	0.3377
	(0.0009)	(0.0000)	(0.0000)
Below poverty pop.	−0.0120	−0.0076	0.0254
	(0.1620)	(0.0000)	(0.0703)
Social orgs count	0.3684	0.1100	0.3514
	(0.0000)	(0.0000)	(0.0000)
Constant	−0.3516	−0.2462	0.8623
	(0.0001)	(0.0000)	(0.0000)
State-level indicators	Yes	Yes	Yes
MSA-level indicators	Yes	Yes	Yes
R^2	.6883	.4945	.6130
N	32,647	32,647	16,658

Note: *p*-Values reported below coefficients.

To overcome this challenge, we complement these visualizations with a multivariate analysis. The main specification is a model that estimates the prevalence of Shopify ventures across the United States. The unit of analysis is the neighborhood (ZCTA), in line with prior work on the spatial distribution of entrepreneurial activity (Sorenson, 2017; Stuart & Sorenson, 2003). Specifically, we estimate the number of unique Shopify ventures across all ZCTAs. The OLS regression utilizes the absolute values of the neighborhood demographic variables, per past work (Uslaner, 1976). Given the nature and distributions of the count variables, we use logged versions of all variables. We further include state-level fixed effects to account for different legal regimes, as well as MSA-level fixed effects to capture variations across metropolitan areas.[15] Standard errors are clustered at the MSA level.

The main result is reported in Model 1 of Table 3, which predicts the logged count of Shopify in the ZCTA. The coefficient of interest, *Black/AA pop.* (*log*), is positive and statistically significant at conventional levels ($b = .031$; $p < .000$). That is, the number of Shopify stores in a given neighborhood is positively associated with the size of the Black population therein. Because both the dependent and independent variables are logged, this coefficient indicates a 1% increase in the prevalence of Black residents in a neighborhood is associated with about

[15]Note that state and MSA dummies can be simultaneously included because these two units of analysis are not geographic subsets of each other, meaning they can (and often do) span each other's boundaries: a state can include multiple MSAs and an MSA can span multiple state boundaries.

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a 3% increase in the number of Shopify ventures. The pattern is observed while controlling for the neighborhood's total population as well as common measures of the level of human capital, financial capital, and social capital as well as state and MSA.

The findings echo our observation that Shopify ventures are commonplace across half of all the neighborhoods. We further seek to uncover whether the pattern is driven by wide use of Shopify across many neighborhoods (a so-called "extensive effect"), or a substantial uptake of Shopify ventures in neighborhoods with at least one Shopify venture (a so-called "intensive effect"). To that end, we present two corollary analyses: (i) an "extensive analyses" of the likelihood of any Shopify ventures across all neighborhoods in the United States, and (ii) an "intensive analysis" which zooms in on those neighborhoods that are home to Shopify ventures (i.e., about half of all ZCTAs) and estimates the number of unique Shopify ventures. The right-hand side retains the aforementioned controls as well as state-level and MSA-level fixed effects.

Model 2 of Table 3 reports our "extensive" analysis, estimating a Linear Probability Model predicting whether a ZCTA had any Shopify ventures; the main coefficient in this specification is positive ($b = .007$; $p = .003$). Next, we report an "intensive" OLS model predicting the logged count of Shopify ventures within a ZCTA (Model 3 of Table 3). This sample is limited to the 16,658 ZCTA that had at least one Shopify venture (representing roughly half of all ZCTAs in the United States). Again, the main coefficient of interest, *Black/AA pop.* (*log*), is positive ($b = .014$; $p < .000$), which reflects an increase of 1.4% in the number of Shopify ventures with a 1% increase in the prevalence of Black residents in a neighborhood.

Taken together, the analyses highlight the role of low-code tools in entrepreneurial inclusion. We find evidence consistent with inclusion not only in terms of geographical spread (as suggested by the extensive analysis) but also in terms of the number of ventures per neighborhood (as suggested by the intensive analysis).

6 | ADDITIONAL ANALYSES

In this section, we present two sets of additional analyses. Section 6.1 facilitates the interpretation of the aforementioned findings by benchmarking them to commonly studied measures of entrepreneurial activity. Section 6.2 describes an analysis that zooms in on areas in the United States with neighborhoods in the top decile of per capita Black population; the section describes the subsample analyses and the full results are available in the Online Appendix.

6.1 | Benchmarking to other forms of entrepreneurial activity

The main analysis presents evidence of the spatial distribution of entrepreneurial ventures. The pattern we documented is indicative of an explanation that low-code tools facilitate entrepreneurial inclusion. Namely, Shopify ventures are more prevalent in neighborhoods that are home to a larger Black population, compared to those that are not. One challenge in interpreting the current set of results is that it gives rise to the following questions: Is the uptick in Shopify ventures in certain neighborhoods unique to low-code based ventures? How does it compare to broader patterns of entrepreneurial activity? Hence, a benchmark against which to assess the pattern of Shopify ventures would be helpful.

We undertake a "book end" approach to inform the interpretation of the main findings. On one side, we explore a "broad benchmark" by documenting the spatial distribution of all

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entrepreneurial activity using a common measure in extant work: a measure of total new ventures activity (Fairlie & Robb, 2008; Fazio et al., 2021; Vedula & Fitza, 2019). To that end, we collect and analyze information for nearly 10 million new businesses all over the United States. However, many of these businesses differ markedly from the e-commerce ventures that we investigate in this study. Therefore, we complement the analysis with a "narrow benchmark" using entrepreneurship data on e-commerce peers. This second analysis leverages data on a few hundred high-growth e-commerce ventures as an alternate benchmark.

This approach offers two informative benchmarks: one that captures all new ventures and another that considers similar peers (Yu & Fleming, 2022). Notably, the volume of Shopify ventures (approximately 160,000) falls in between these two bookends (i.e., the broad benchmark captures a few million ventures and the narrow benchmark a few hundred).

6.1.1 | Broad benchmark: All entrepreneurial ventures

The first benchmark concerns all new ventures registered in the focal neighborhood. We are motivated by a large body of work using new businesses as an indicator of entrepreneurship. This measure of entrepreneurial activity aggregates businesses in each area (e.g., a neighborhood) irrespective of industry. It is a common approach in studies of entrepreneurship, in general (Fazio et al., 2021; Sorenson et al., 2016), and entrepreneurial access, in particular (Fairlie & Robb, 2008; Gagliardi & Sorenson, 2023; Vedula & Fitza, 2019).

We draw on the Startup Carography database to create a broad measure of entrepreneurial activity (Andrews et al., 2022). The data are based on detailed business registration records in almost all US states through 2016, and has been used in recent studies (Denes et al., 2023; Fazio et al., 2021; Yu & Fleming, 2022). Our measure, *Entrepreneurial Businesses*, captures the total number of new businesses registered in a given ZCTA over the most recent five years of data.[16] As expected from a broad benchmark, we observe some entrepreneurial activity (i.e., nonzero value) in 30,951 (95%) of all ZCTAs. Following the main analysis, we use a logarithm of this measure, plus one, as the dependent variable.

Model 1 of Table 4 replicates the specification of the main model using the new measure. It reports an OLS estimation of *Entrepreneurial Businesses*. The coefficient of interest, *Black/AA pop*. (*log*), is negative and statistically does not differ from zero at conventional levels ($b = -.003$; $p = .447$). That is, there is evidence of a negligible or negative association between the number of new businesses in a given neighborhood and the size of the Black population therein. As one might expect, the total size of the population features a positive coefficient; more populated neighborhoods have more new businesses overall. The measures of human, financial, and social capital exhibit similar coefficient size and magnitude. Notably, the *R*-squared stands at a high level of .92, suggesting that this traditional measure of overall entrepreneurial activity is indeed strongly associated with well-known measures of neighborhoods' capital endowments and population size.

The findings offer a useful benchmark against which to interpret the findings of the main analysis. The main finding that Shopify ventures are more common in neighborhoods with a larger Black population stands in contrast to the weakly negative pattern we document for traditional ventures. We subsequently check whether the difference is of statistical significance. To that end, we jointly ran the current model and the main Shopify model using a Seemingly

[16]The results are robust to aggregating new business numbers over different periods; either 3 years or 10 years.

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TABLE 4 "Book ended" benchmark analyses.

Benchmark	(1) Broad	(2) Narrow
Dependent variable	**Log (All new businesses +1)**	**Log (VC-backed e-commerce ventures +1)**
Black/AA pop.	−0.0029	−0.0002
	(0.4468)	(0.5636)
Total pop.	0.1695	−0.0074
	(0.0087)	(0.0594)
Bachelor degrees pop.	0.3432	0.0029
	(0.0000)	(0.0652)
Below poverty pop.	−0.0129	0.0012
	(0.3437)	(0.0169)
Social orgs count	0.4950	0.0062
	(0.0000)	(0.0796)
Constant	−1.2578	0.0232
	(0.0000)	(0.0894)
State-level indicators	Yes	Yes
MSA-level indicators	Yes	Yes
R^2	.9175	.0495
N	32,647	32,647

Note: p-Values reported below coefficients.

Unrelated Regression (SUR) estimation; the difference between the two coefficients on *Black/AA pop*. (*log*) is highly significant at conventional levels ($\chi^2 = 27.31$; $p < .000$). In light of this observation, there is further confidence in the interpretation of low-code tools as facilitating entrepreneurial inclusion. It is noteworthy that total population size is positively associated with traditional measures of entrepreneurial activity yet exhibits a negative association with Shopify ventures. The opposite effect likely reflects the fact that the current benchmark, *Entrepreneurial Businesses*, captures all ventures irrespective of industry.

6.1.2 | Benchmark 2: E-commerce ventures

As we contemplate an appropriate benchmark, we shift from a measure of overall entrepreneurial activity commonly used in the literature and consider the features of Shopify businesses as delineated in Section 2. These are e-commerce businesses that share many features with well-known e-commerce businesses launched over the past decades. One feature of e-commerce businesses—irrespective of whether launched using low-code tools or not—is that they often have broad geographical reach. These Hi-Tech ventures are less sensitive to the magnitude of the local demand (e.g., population size) because they can leverage the Internet to cater to consumers across a broader region or even internationally. Consider for example, Denver-based Orion Brown talking about the customer base of her Shopify venture, Black Box Travel; "*My community is made up of millions of African-American travelers who love travel as a form of*

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self-care and are looking for their tribe."[17] The potential for broad reach underlies another feature shared by all e-commerce businesses; that is, the opportunity to leverage reach to drive ventures' growth. There are notable examples to that effect among VC-backed e-commerce ventures (e.g., Bonobos, Dollar Shave club, Warby Parker), as well as Shopify based e-commerce ventures (e.g., Butcher Box, Figs, or Black Travel Box).

Accordingly, our second benchmark focuses on e-commerce peers. Namely, e-commerce ventures that are not built using low-code tools, yet share the aforementioned features (i.e., broad reach and growth potential). Per our "book end" approach, the former benchmark employed an expansive measure of all entrepreneurial ventures (irrespective of industry or type), whereas the current benchmark utilizes a narrow set of e-commerce ventures commonly viewed as Hi-Tech high-growth ventures (Andrews et al., 2022; Hallen, 2008).

We undertake the following steps to create the second benchmark, *VC E-Commerce Businesses*. First, we identify the population of e-commerce businesses documented by Pitchbook, a common source of information for growth-oriented businesses (Gornall & Strebulaev, 2020; Lerner & Nanda, 2020). This resulted in 369 e-commerce businesses that were active and venture-backed during the 2010s and had address data that could be matched to ZCTAs in our sample. Next, we cross-check with our sample to remove those ventures that were based on Shopify. About 141 ventures (38%) dropped as a result, leaving us with 228 e-commerce ventures. Finally, we constructed *E-Commerce Businesses* as the total number of non-Shopify e-commerce businesses in a given ZCTA. The dependent variable is the logarithm value of this measure plus one.

Before we turn to the regression analysis of this second benchmark, we note the total count of these ventures. Only 228 US-based e-commerce ventures are recorded by Pitchbook, a comprehensive source of information for Hi-Tech growth businesses. It is about 700-fold smaller than the 159,663 US-based Shopify ventures in our sample. Comparing the total number of e-commerce ventures (228 VC-backed vs. 159,663 Shopify-based) offers an early indication of the role of low-code tools in entrepreneurial access. It echoes the discussion in Section 2 and Levie's observation that Shopify dramatically lowers the barriers to entrepreneurship. Of course, the comparison should be viewed with caution because we know that the number of VC-backed ventures is a small fraction of all entrepreneurial ventures in the United States. The comparison is still informative, however, especially if we recall that even among the original list of VC-backed e-commerce ventures of 369, a large minority (141, or 38%) were Shopify ventures.

Model 2 of Table 4 replicates the main model specification, regressing the log *VC E-Commerce Businesses* on a set of variables used in the main analysis. One challenge of this model is associated with the scarcity of observations; the dependent variable takes nonzero values for only 162 (0.5%) of all ZCTAs. Consequently, we expect the coefficients to exhibit lower magnitude and statistical significance. This is also evident from the lower total fit of the model; an R-squared value of .049. As for the coefficient of interest, *Black/AA pop*. (*log*), it is close to zero and statistically insignificant ($b = .000$; $p = .564$). In contrast to Shopify-based e-commerce ventures that are more common in neighborhoods with a larger Black population, we observe no meaningful relationship for other e-commerce ventures. Again, we jointly ran this model and the main Shopify model in a SUR to confirm the difference between the two coefficients on *Black/AA pop*. (*log*); the difference is substantial and statistically significant ($\chi^2 = 32.17$; $p < .000$). This observation offers further confidence in the interpretation of low-code tools as facilitating entrepreneurial inclusion. The coefficients on the other variables are of magnitude and significance similar to that in the main analysis, with the exception of the financial capital proxy.

[17]https://www.shopify.com/uk/blog/how-these-black-and-indigenous-merchants-are-empowering-their-communities. See also, https://www.oprahdaily.com/beauty/a44520546/orion-brown-black-travel-box/.

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In sum, the benchmark analyses offer instructive guidance. While one may argue with the nature of either of the two benchmarks, they are aligned with extant work and can be viewed as "book ending" the focus of our investigation. Along these lines, the 159,663 Shopify ventures fall between the first "broad" benchmark of nearly 10 million entrepreneurial ventures (irrespective of industry or type) and the second "narrow" benchmark of 228 VC-backed e-commerce ventures. Looking across the models, we observe that neighborhoods with a larger Black population exhibit a nonpositive association with traditional measures of entrepreneurial activity (i.e., our two benchmark measures), which further underscores the unique pattern we do observe for Shopify ventures (i.e., a positive and significant association).

6.2 | Focusing on areas with the highest fraction of Black population

We complement the US-wide patterns reported above with additional analyses that zoom in on areas with the highest per capita Black population. The subsample analysis focuses on variation in entrepreneurial activity across neighborhoods in the top decile of per capita Black population and those immediately adjacent to them. These neighborhoods are disproportionately in the southern area of the United States (see Figure A.1). The area covered by the subsample consists of 7277 ZCTAs.[18] Below, we describe three key observations and report the detailed results in Table A.1 of the Online Appendix.

First, within the subsample, the mean number of Shopify-based ventures stands at 6.94. That is about 40% above the mean value for the full sample. In other words, Shopify-based ventures are more common in these areas. Other neighborhood characteristics are also unique to those parts of the United States; for example, the mean number of population under poverty and count of social organizations are double that for the whole of the United States, and their standard deviation values are higher as well. It follows that the patterns reported below should be interpreted as applicable to specific areas and need not reflect the US-wide coefficients.

Second, we reran the models from Tables 3 and 4 for the subsample. The coefficients of interest differ from the results above. Consider, for example, the coefficient of *Black/AA pop*. (*log*) in the Shopify analysis (Model A.1–1) which is now negative and does not differ from zero at conventional levels. In the analysis of all new business registrations (Model A.1–2), the coefficient on *Black/AA pop*. (*log*) is not only negative but also highly statistically significant.

The third observation, however, is that despite differences in sign and magnitude in each of the models of Table A.1, the relative effect across these models is consistent with the prior benchmark analyses. In other words, even within the subsample areas, we continue to find a stronger association between the size of the Black population and Shopify-ventures, relative to its association with other forms of entrepreneurial activity.

7 | DISCUSSION AND CONCLUSION

This study examines a novel phenomenon of low code e-commerce entrepreneurial activity. Specifically, we investigate entrepreneurial inclusion through the lens of the geographic distribution of (Shopify-based) ventures in the United States. We observe substantial cross-

[18]The subsample covers ZCTAs with the highest per capita Black population (nationally) and any ZCTA with a shared border. See the Online Appendix for additional detail.

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neighborhood variation in the number of these e-commerce ventures, controlling for the state and MSA in which they are located. We find that the number of Shopify ventures features a uniquely positive association with the number of Black residents in a neighborhood. To interpret the results, we further examine the spatial distribution of entrepreneurial activity using two other measures from the literature. The benchmark analyses further reveal that Shopify ventures exhibit a uniquely stronger association between the number of ventures and the Black population.

Overall, these patterns suggest that low-code tools reshape entrepreneurial access. Our abductive approach finds that traditional explanations of entrepreneurial activity persist, yet do not fully explain the distinct patterns of Shopify ventures. In the following subsections, we first elaborate on how our findings can be interpreted in light of related mechanisms from prior entrepreneurship research (Section 7.1) and then expand on how future research might go about testing the novel implications of these mechanisms in the context of low-code entrepreneurship (Section 7.2). Section 7.3 concludes.

7.1 | Discussing theoretical mechanisms

As a low code tool, Shopify lowers the barriers to entrepreneurship, which is advantageous to many and particularly to those who have been traditionally underrepresented in entrepreneurship. The "book end" benchmarking to other forms of entrepreneurial activity further illuminates the unique pattern of entrepreneurial inclusion exhibited by Shopify ventures. These findings speak to two mechanisms in the literature; the first relates to who pursues entrepreneurship, and the second concerns the entrepreneurial opportunities being pursued.

7.1.1 | Entrepreneurial access and "who" pursues entrepreneurship

Our findings engage with prior studies of inclusion and who pursues entrepreneurship. This literature is motivated by the well-documented Black/White racial divide in representation in entrepreneurship (Bruton et al., 2023); be that in terms of self-employment and business ownership (Fairlie & Robb, 2008; Fairlie & Robinson, 2022) or Hi-Tech entrepreneurship (Fairlie & Robinson, 2022; Gompers & Wang, 2017). These racial disparities in entrepreneurship are observed for individuals and communities across the United States and cannot be fully explained by the endowment of financial, human, or social capital (Fairchild, 2008; Fischer & Massey, 2000).

Recent work finds that barriers to entrepreneurship, and specifically Black entrepreneurship, can be mitigated through different mechanisms. Extant work uncovers the role of (i) top-down policies and (ii) bottom-up training as effective in facilitating entrepreneurship among underrepresented communities (Assenova, 2020; Chatterji & Seamans, 2012; Lyons & Zhang, 2017a; Pongeluppe, 2024; Sørensen & Sharkey, 2014). As we review below, common to these mechanisms is that they not only decrease the barriers to entrepreneurship but also have a relatively greater impact on Black entrepreneurs. By documenting the role of technology (i.e., a low-code tool for e-commerce), we join these investigations of entrepreneurial access.

Consider the evidence regarding policy as a mechanism of entrepreneurial access. Extant work reports that policy affects the cost of entrepreneurship, and that it often has a disparate impact on previously underrepresented populations; that is, "the decision to become an

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entrepreneur is linked to the structure of inequality in existing organizations, and variation across opportunity structures influences differences in the rate of entrepreneurship" (Sørensen & Sharkey, 2014). We know that policies that decrease the cost of entrepreneurship (or relatedly, change the ability to be in employment) have disparate impact on Black populations. For example, credit card deregulation during the 1980s reduced financing costs leading to an increase in entrepreneurial entry. Notably, the effect was particularly strong for Black entrepreneurs, and further intensified for communities in states with a history of racial discrimination (Chatterji & Seamans, 2012).

Similarly, training programs and accelerators play an important role in facilitating entrepreneurship. Extant evidence reveals that these programs encourage entrepreneurship by enhancing financial, human, or social capital (Assenova, 2020; Hallen et al., 2020; Pongeluppe, 2024). Notably, there is evidence that the effects can be particularly strong for entrepreneurs from underrepresented groups. For example, Lyons and Zhang (2017b) find that non-Caucasians engage with entrepreneurship more than Caucasians after partaking in a technology entrepreneurship training program.

To the extent that low-code technology constitutes a distinct mechanism, it joins the policy and training mechanisms in contributing to our understanding of entrepreneurial inclusion. Common to these mechanisms is that they reduce the hurdles to launching a new venture. For scholars of entrepreneurial access, the evidence from the aforementioned works and this study highlights a critical observation. Not only do policy, training, or low-code technology affect the barriers to entrepreneurship, but also these mechanisms have a distinct effect for traditionally underrepresented individuals and communities.

7.1.2 | Entrepreneurial access and "what" opportunities are being pursued

Our findings can be viewed and informed by work on entrepreneurial opportunities. Extant work suggests that opportunities are discovered by certain individuals given their idiosyncratic background and experiences (Kirzner, 1979; Shane, 2000). However, we know that launching a venture necessitates financial, human, and social capital (Hallen, 2008; Hallen et al., 2014; Parker, 2018). Informed by the well-documented Black/White racial divide (Bruton et al., 2022; Fairlie & Robinson, 2022; Younkin & Kuppuswamy, 2017), we recognize that individuals from certain demographics might have been able to see, yet not seize, entrepreneurial opportunities. The advent of low-code tools empowers those individuals not only to discover an opportunity but also to exploit it. Consider the account of *Black Girls Run!* Co-founders, Ashley Hicks-Rocha and Toni Carey[19];

> During their first year of hitting the pavement together, the women realized that not only were they the only black women participating in the local running groups and road races – they were also the only people of color, period... The isolation they felt in a sport they loved, along with a history of diabetes and other health problems in both of the women's families, prompted Toni and Ashley to share their story... "When we decided to launch the running groups, we saw an opportunity to start monetizing the business. It just went viral..." [as of 2016] Black Girls RUN! has a membership of over 150,000 and hosts over 70 running groups across the

[19]https://www.shopify.com/uk/blog/195184457-meet-the-founders-of-black-girls-run.

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country. Through their online store, they've funded their dream of impacting obesity rates in America, selling Black Girls RUN! merch to women proud to represent the community that changed their lives.

One implication of this is that low-code tools also alter the scale of opportunities that are deemed worth pursuing. A parallel observation is shared in recent strategy work on the rise of "middle tail" strategies (Benner & Waldfogel, 2023). The underlying theoretical argument is that digital tools not only enable producers to seize and profit from the "long tail" of existing products and services, but also impel the creation of new products and services. Put differently, the lower barriers to production and distribution result in a qualitatively distinct strategy. Benner and Waldfogel (2023) find that digital tools facilitate the pursuit of products catering to moderately sized markets that were previously deemed too small. Moreover, they report that those who see and seize such opportunities are predominantly new ventures.

This mechanism aligns and informs our observations. When launching an e-commerce venture was a costly endeavor, only opportunities of a substantial magnitude marshaled the investment necessary to take them forward (Drover et al., 2017). Opportunities to address smaller, moderately sized markets might have been discovered yet failed to attract VC funding. Indeed, the e-commerce ventures of the 1990s and 2000s addressed mainstream markets where the total addressable market was large. The advent of low-code tools may encourage a "middle tail" strategy (Benner & Waldfogel, 2023) in e-commerce entrepreneurship. Low-code tools imply that a broader set of markets represent profitable entrepreneurial opportunities; they might have a lower total addressable market yet represent a lucrative opportunity in light of the lower required investment (Hallen & Hallen, 2024).

7.2 | Future opportunities to unpack theoretical mechanisms

The nature of our large-scale data allowed us to study a type of entrepreneurship that is increasingly prevalent across the United States. By analogy to the burgeoning accelerator literature, we emphasize that the careful scholarly exploration of a novel phenomenon can advance our understanding of entrepreneurial activity (Benner & Waldfogel, 2023; Clough et al., 2019; Hallen et al., 2020). Building on O'Mahony and Cohen (2022), we hope this study—and Section 2 specifically—presents definitions and shared language for talking about the phenomenon of low-code-based ventures. Future work can further unpack the mechanisms that underlie this novel phenomenon and advance our overall understanding of entrepreneurial activity. The preceding section highlights two potentially related mechanisms—related to "who" is pursuing an opportunity as well as "what" opportunity is being pursued—that we believe are consistent with our empirical results and present promising paths for research.

Future work focused on the "who" mechanism should investigate the decisions undertaken by individuals who opt to pursue low-code entrepreneurship. This line of investigation could shed light on the unique hurdles and opportunities faced by the entrepreneurs who created these ventures. Exploring the decision-making process that leads up to the creation of the Shopify-based ventures, we observed calls for capturing not only the "path taken" (i.e., the launch of a Shopify venture) but also the "paths not taken." Admittedly, our large sample analysis is well-suited to identify the patterns of the "(Shopify) path taken," yet it cannot shed light on the counterfactuals that each and every Shopify entrepreneur faced. Alternative paths may have been "desired yet unavailable" (e.g., previously, launching an e-commerce venture was

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mainly the prerogative of individuals with technical expertise) or alternative paths may have been "available yet foregone" (e.g., previously, these individuals would have launched mainly locally focused small-scale ventures). Drawing parallels to the recent of stream of accelerator work, we highlight the opportunity to harness qualitative methods to inform us of the alternative paths that were considered by the founders of such ventures (Hallen et al., 2020; O'Mahony & Cohen, 2022).

Future work focused on the "what" mechanism might aim to document the specific nature of the opportunities that are being pursued via low-code entrepreneurship. For example, what does the distribution of potential market size look like for these ventures? Specifically, is it that the unique cost and expertise profile associated with low-code tools enables "middle tail" ventures (Benner & Waldfogel, 2023; Hallen & Hallen, 2024)? Addressing these questions can shed new light on the pursuit of entrepreneurial opportunities. Opportunities that were previously recognized yet financially infeasible may become viable via low-code entrepreneurship. That is, low-code tools may expand entrepreneurial access in the sense that they may give rise to a new and distinct class of venture sizes. If so, a related question concerns what dimensions the products and services developed via low-code entrepreneurship differ from other forms of entrepreneurship.

Tackling this question can inform us if low code tools expand entrepreneurial access and specifically entrepreneurial inclusion. To see that, recall the anecdotal evidence presented earlier in the paper. Neither Orion Brown (founder of *Black Box Travel*), nor Ashley Hicks-Rocha and Toni Carey (co-founders of *Black Girls Run!*) set up their businesses as a small-scale local venture. And while these ventures are not of small scale, they do not fit the hyper-scale VC-backed "unicorn" template that has received substantial attention in recent entrepreneurship studies. It suggests that entrepreneurs utilizing low code tools are distinct not only in the size of the market opportunity their ventures address but also in the nature of the opportunity they uncovered. These entrepreneurs may recognize market niches that were "desired yet unavailable" previously to individuals of similar profile. Future work could explore whether Shopify ventures launched by individuals from underrepresented communities (e.g., Black entrepreneurs) are more likely than other ventures to deliver products and services that target their communities (e.g., Black customers).

Finally, we believe there is an opportunity to go beyond Shopify and document different low-code technologies and their impact on entrepreneurial access. Another natural avenue for research involves attempting to understand the performance outcomes of using low-code tools. However, we caution that interpreting performance outcomes in light of other forms of entrepreneurship may be challenging, as the phenomenon we have outlined in this article is likely related to observed survival and failure rates in interesting ways (Dushnitsky & Stroube, 2021). It will require careful scholarship to both collect meaningful performance data and accurately interpret it vis-à-vis existing performance findings in the entrepreneurship literature.

7.3 | Conclusion

To the extent that low-code tools alter the role of human and financial capital, there is a need to rethink current orthodoxies in the study of entrepreneurship. For example, the prevailing focus on institutional-economic interventions as drivers of entrepreneurship (e.g., policy aimed at stimulating entrepreneurial activity, in general, or entrepreneurship by underrepresented

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groups, in particular) may be unnecessarily limited. In fact, our analysis of Shopify ventures points to an alternative techno-economic approach to expanding entrepreneurial access.

Moreover, extant work often exhibits a binary view of entrepreneurship; focusing either on self-employment and local businesses, or VC-scale ventures addressing major markets (i.e., "Unicorns"). We join a conversation motivated by recent work (Benner & Waldfogel, 2023; Hallen & Hallen, 2024), and underscore the prospect of future work that can explore another type of entrepreneurship; that is, one that may be characterized not as self-employment nor as VC-scale opportunities, but rather as something in the middle of these extremes. We believe that investigating the entrepreneurial activity unlocked by low-code tools can advance our understanding of entrepreneurial access not only by those from underrepresented communities, but also those who pursue previously unattainable "middle tail" opportunities.

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DATA AVAILABILITY STATEMENT

The data and code that support the findings are available at the following Open Science Framework link: https://osf.io/4gpyr/.

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